

Ibnu Cipta Ramadhan

Edge AI | Computer Vision | IoT Engineer

✉ ibnucr@email.com [ibdhan.github.io](https://github.com/ibdhan) [ibdhan](https://github.com/ibdhan) [ibdhan](https://www.linkedin.com/in/ibdhan) [ibdhan](https://www.linkedin.com/in/ibdhan)
📍 Surabaya, Indonesia (Available for Remote Collaboration)

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Professional Summary

Embedded AI, IoT, and Computer Vision Engineer with over five years of experience developing intelligent edge systems, from embedded hardware and sensor integration to real-time AI deployment. Experienced in building production IoT devices, edge-to-cloud solutions, robotics perception systems, and computer vision applications using NVIDIA Jetson, ROS 2, and PyTorch. Comfortable working across the full engineering stack, from electronics and firmware to AI optimization, device integration, and scalable IoT system development.

Skills

AI / CV	2D and 3D Object Detection, YOLO, OpenCV, PyTorch, TensorRT
Edge / Embedded	NVIDIA Jetson (Orin, Xavier), CUDA, ROS 2, LiDAR
IoT	GPS/Fleet Tracking, MQTT, LoRa, Telemetry, Redash, Metabase
Programming	Python, C, C++, SQL, Bash, LaTeX
Development	Docker, Git, Agile, Scrum, AWS

Work Experience

University Lecturer	2024.08 – Present
Politeknik Elektronika Negeri Surabaya (PENS)	Surabaya, Indonesia

- Conduct research in edge AI systems in computer vision, robotics, and embedded intelligence.
- Supervised undergraduate engineering projects involving AI, robotics, and IoT deployment.
- Designed laboratory modules covering sensor & actuators, instrumentation, and digital signal processing.

Senior Hardware & Internet of Things (IoT) Engineer	2024.02 – 2024.07
McEasy – PT. Otto Menara Globalindo	Surabaya & Jakarta, Indonesia

- Designed end-to-end IoT solutions for logistics and fleet management applications.
- Worked directly with enterprise customers to translate operational requirements into deployable hardware and software solutions.
- Collaborated with cross-functional software teams to integrate embedded devices into cloud platforms.
- Supported field deployment, debugging, and hardware validation.

Hardware & Internet of Things (IoT) Engineer	2023.02 – 2024.01
McEasy – PT. Otto Menara Globalindo	Surabaya & Jakarta, Indonesia

- Integrated GPS tracking devices and telematics sensors with fleet monitoring platforms.
- Designed and executed test cases for GPS trackers, sensors, and embedded hardware before production deployment.
- Analyzed telemetry data to investigate product issues, evaluate device performance, and support data-driven feature development.
- Developed hardware and sensor-related features based on customer and product requirements.

Research Assistant - Collision Avoidance System Development For Mining	2023.02 – 2023.05
STEI – ITB	Bandung, Indonesia

- Developed perception and sensing modules for collision avoidance systems in mining vehicles.
- Integrated LiDAR, camera, and sensor data for environment perception and obstacle detection.
- Evaluated system performance through simulation and field testing.

Research Assistant - Autonomous Tram Development Using AI

STEI – ITB

2021.01 – 2023.04

Bandung, Indonesia

- Developed AI-based perception systems for autonomous tram navigation using LiDAR data.
- Implemented and evaluated 3D object detection algorithms on NVIDIA DRIVE AGX platforms.
- Collected, processed, and analyzed real-world datasets for model development and validation.

Internet of Things (IoT) Engineer

Mceasy – PT. Otto Menara Globalindo

2018.02 – 2020.08

Surabaya & Jakarta, Indonesia

- Designed embedded hardware for GPS tracking devices.
- Integrated GNSS, cellular communication, accelerometers, and digital I/O.
- Performed hardware bring-up, firmware validation, and manufacturing support.
- Worked closely with firmware and backend teams for production deployment.

Internet of Things (IoT) Engineer (Part-time)

Mceasy – PT. Otto Menara Globalindo

2017.04 – 2018.02

Surabaya & Jakarta, Indonesia

- Designed embedded hardware for GPS tracking devices.
- Integrated GNSS, cellular communication, accelerometers, and digital I/O.
- Performed hardware bring-up, firmware validation, and manufacturing support.
- Worked closely with firmware and backend teams for production deployment.

Internship

PT. Pertamina Hulu Energi – West Madura Offshore

2015.08 – 2015.09

Gresik, Indonesia

- Assisted engineers in maintenance and inspection activities for offshore production facilities.
- Supported data collection and technical documentation for engineering operations.

Projects

LiDAR-Based 3D Perception for Autonomous Tram

2021 – 2023

Tech: PyTorch • CenterPoint • LiDAR • NVIDIA DRIVE AGX Pegasus

- Developed a LiDAR-based 3D perception pipeline for autonomous tram navigation.
- Collected, processed, and annotated large-scale point cloud datasets for model training and evaluation.
- Deployed on NVIDIA DRIVE AGX Pegasus; achieved 20 FPS on real-world point cloud data
- Demo: <https://www.youtube.com/watch?v=Dft3OKFTdvs>

GPS Fleet Tracking Platform

2023 – 2024

Tech: GPS • LTE • IoT • Embedded Systems • Sensor Integration

- Integrated GPS tracking devices and telematics sensors with cloud-based fleet management platforms.
- Designed validation procedures and test cases for GPS trackers and embedded hardware.
- Developed hardware and sensor-related features to improve system functionality.

Edge AI Deployment on NVIDIA Jetson

2025 – 2026

Tech: Jetson Xavier NX • Jetson Orin Nano • TensorRT • CUDA • Docker

- Deployed and optimized deep learning models on NVIDIA Jetson edge devices.
- Benchmarked inference performance using TensorRT on multiple Jetson platforms.
- Developed containerized deployment workflows using Docker for reproducible development.
- Repo: <https://github.com/ibdhhan/ros2-yolo>

Real-Time Indonesian Traffic Sign Detection

2025

Tech: YOLO • PyTorch • OpenCV • Computer Vision

- Developed an object detection model for Indonesian traffic sign recognition using YOLO.
- Evaluated transfer learning and training-from-scratch approaches on a custom dataset.
- Optimized model performance for real-time edge deployment.
- Repo: <https://github.com/ibdhhan/yolov11-transfer-learning>

IoT Telematics Monitoring Platform

2023 – 2024

Tech: Python • PostgreSQL • Redash • Metabase • IoT

- Developed data processing pipelines for GPS and telematics sensor data.
- Built monitoring dashboards and analytical tools for fleet performance and device health.
- Analyzed telemetry data to support feature development and operational decision-making.

Education

Master of Science (M.Sc.) – Electrical Engineering

2020 – 2023

Institut Teknologi Bandung (ITB)

Bandung, Indonesia

- **Thesis:** LiDAR-Based 3-Dimensional Object Detection for Autonomous Tram Perception System

Bachelor of Applied Science (B.ASc.) – Electronics Engineering

2016 – 2018

Politeknik Elektronika Negeri Surabaya (PENS)

Surabaya, Indonesia

- **Thesis:** Binaural Sound Source Localization System Based on Fast Fourier Transform Using Interaural Phase Difference as Cue

Diploma in Electronics Engineering

2013 – 2016

Politeknik Elektronika Negeri Surabaya (PENS)

Surabaya, Indonesia

- **Thesis:** Gas Monitoring System Oxygen, Carbon Monoxide, and Carbon Dioxide, with Integration on Wearable Device

Languages

English Professional working proficiency

Bahasa Indonesia Native proficiency